

Your grade will be notified, and you will be able to access the feedback to your answers,

at 7:00pm of Thursday, June 10, 2021.

1.

In the closed economy Macrolandia, there are only one-year and two-year bonds. The current and the future expected inflation rates are both zero, so that the real and the nominal interest rates are equal. In period $t + 1$ ("the future"), a standard IS-LM model describes the functioning of the economy. In particular, in the year $t + 1$ consumption depends only on disposable income at $t + 1$, and investment on the interest rate and output at $t + 1$ only. Turning now to the current period (t , "the present"), consumption is increasing in both current and expected future disposable income, $C_t = C(Y_t - \bar{T}_t, Y_{t+1}^e - \bar{T}_{t+1}^e)$, investment depends positively on current and expected future income levels, and negatively on the current and expected future levels of short-term interest rates, the demand for money has the usual functional form, and G and T are exogenous as usual. Starting from an initial equilibrium, at time t the government of Macrolandia announces an equal decrease in net taxes in both periods, $\Delta \bar{T}_t = \Delta \bar{T}_{t+1} < 0$. At the same time, the central bank announces that it will adjust the policy rate so as to prevent any change in equilibrium income that, at time t - and at time t only - could be caused by the fiscal policy just described. Explain how these announcements, which are unexpected and credible, affect the time t equilibrium levels of income, consumption, investment and national saving in Macrolandia.

SKETCH OF ANSWER

At $t + 1$, the decrease in net taxes will shift to the right the IS curve for that period, thus leading to an increase in equilibrium income. At time t , the IS curve will shift to the right both because of the cut in current and future expected net taxes (something that boosts current consumption), and due to the rise in future expected income (a rise that leads to higher time t consumption and investment). It follows that, to prevent Y_t from increasing, the central bank will have to raise the policy rate. In the new equilibrium that will prevail at time t , consumption will be higher (Y_t unchanged, but $\bar{T}_t$ and $\bar{T}_{t+1}^e$ lower, and Y_{t+1}^e higher), G unchanged, and I_t lower (since the supply of goods is unchanged, the same must be true about the demand for goods, something that – with a higher C_t and an unchanged G – is only possible if investment falls by the same amount by which consumption has gone up). Finally, national saving, in equilibrium equal to investment, will have fallen by the same amount by which I_t has gone down.

2.

With reference to the same economy studied in the previous question, do you think that the permanent decrease in net taxes and the central bank intervention described before will lead to a change in the time t yield to maturity on the two-year bonds circulating in Macrolandia, i_{2t} ? If no, why? If yes, in which direction will i_{2t} change, and by how much? Finally, assuming for simplicity that at time t the yield curve was initially horizontal, explain how the position and the slope of that curve will change after the fiscal policy and monetary policy announcements studied above.

SKETCH OF ANSWER

Since $i_{1,t+1}^e$ does not change and i_{1t} goes up, i_{2t} – approximately equal to the average of the current and the future expected short-term interest rates – will rise. More precisely, it will go up by half of the increase in the current short rate [$\Delta i_{2t} = \frac{1}{2}(\Delta i_{1t} + \Delta i_{1,t+1}^e) = \frac{1}{2} \Delta i_{1t}$, being $\Delta i_{1,t+1}^e = 0$]. Finally, the new yield curve will lie above the old one and will be negatively sloped, as its first point – that, in the diagram where the yield curve is drawn, represents the value of the current short rate, i_{1t} – will rise above the initial one by more than its second one, that is the value that i_{2t} takes on.

3.

Consider an open economy under a flexible exchange rate regime. Domestic and foreign prices are constant and for simplicity both equal to 1 ($P = P^* = 1$), consumption depends positively on disposable income, investment is entirely exogenous ($I = \bar{I}$), net exports depend positively on foreign income, negatively on domestic income, and positively on the real exchange rate, so that the Marshall-Lerner condition is not met. Finally, G and T are exogenous, and the country's central bank chooses the domestic interest rate. Having discussed if, and why, the IS curve of the "IS-LM-interest parity condition" model that describes this economy will have a negative, zero or positive slope, suppose that the government raises $\bar{G}$. To prevent this fiscal policy intervention from affecting the equilibrium level of income, the central bank intervenes by changing the policy rate. To make sure that equilibrium income will remain constant, should the central bank increase or decrease the policy rate? Following these fiscal and monetary policy interventions, which of the three curves (the IS, the LM, and that representing the interest parity condition) will shift in the plane where they are drawn, and in which direction? Finally, discuss how the exchange rate, money supply, money demand, consumption, investment, government purchases of goods and services, and net exports will have changed in the new equilibrium that the economy will reach. Explain.

SKETCH OF ANSWER

The IS will have a positive slope. In fact, if – starting from a goods market equilibrium position – the interest rate goes up, investment does not change (it is exogenous), the exchange rate will appreciate (via interest parity condition) and net exports will increase, since they depend positively on the real exchange. To go back to goods market equilibrium, the supply of goods will have to rise. It follows that goods market equilibria with higher levels of i are in this case associated with higher levels of Y – the IS curve will be positively sloped in the (Y, i) plane.

The increase in $\bar{G}$ shifts the IS to the right. To prevent income from rising, the central bank will have to cut the policy rate, thus shifting downwards the LM curve until it crosses the new IS for the initial value of Y . In fact, in this economy a decrease in the interest rate depreciates the exchange rate and lowers net exports, and with them the demand for goods. The curve that represents the interest parity condition will not shift following these fiscal and monetary policy interventions (its position in the plane depends on E^e and i^ , which have not changed).*

In the new equilibrium, E will be lower (because i is lower), money supply M higher (an increase needed to decrease the policy rate), money demand higher by the same amount by which M has gone up (i is lower, Y unchanged), consumption unchanged (like disposable income), investment unchanged (it is exogenous), net exports smaller (due to the exchange rate depreciation), and $\bar{G}$ of course higher. Being Y unchanged, one can also conclude that net exports will have gone down by the same amount by which $\bar{G}$ has been raised.

4.

By making explicit reference to the government budget constraint – and, should you find it helpful, also to the diagram employed to perform a graphical analysis of the evolution of the debt-to-GDP ratio over time –, explain if and why you agree, or do not agree, with the following statement:

“In a country where $r > g$ and that inherits from the previous period a stock of government debt $B_{t-1} > 0$, the debt-to-GDP ratio will keep growing over time, and this irrespective of the value – by assumption, constant – that d , the ratio of the primary deficit to GDP, takes on”.

SKETCH OF ANSWER

The statement is incorrect.

In fact, if the government runs a primary surplus ($d < 0$), the debt ratio could end up falling over time even if $r > g$. This would be the case if, for a given debt ratio b_{t-1} inherited from the past, the primary surplus – that tends to lead to a lower b – is large enough to more than offset the impact on the debt ratio of the fact that r is greater than g , or alternatively if, for a given $d < 0$, b_{t-1} is small enough.